



Human Headphones, the world's first true wireless over-ear headphones

nials and Gen Zs consider audio as a vital escape from the overload of visual stimulation today. According to Harman, the demand for personalised audio experiences is growing and will continue to grow exponentially. Not just headphones, but Harman also stated that the category of smart speakers is one of the fastest growing tech sectors in the US with nearly one-third of American households having one. We see the smart speakers category penetrating the Indian market more in the coming years as well. It's safe to say that the innovation in the audio gadgets industry hasn't peaked, and the next 20 years will possibly see myriad inventions and changes that continue to revitalise the industry. Let's look at a few ways the audio gadgets industry will possibly change in the next 20 years.

New form factors

Over the last couple of years, we've already seen interesting, new form factors of headphones in the audio industry. Two examples that immediately come to mind are the Bose Frames AR Audio Sunglasses and the Human Headphones, both of which launched in 2019. The Bose Frames are a pair of premium sunglasses that are also a pair of wireless headphones with AR (augmented reality) capabilities. The AR capabilities are, however, audio only and the

idea didn't really take off causing the company to shut down its AR division. Nevertheless, it is a pretty intriguing idea that can be game-changing if implemented well. The Human Headphones is a strange over-ear true wireless headphone that has no headband and sits snugly around your ears instead. This could be a potential game changer as well since it mixes the portability of true wireless earphones with the enhanced noise isolation

capabilities of over-ear headphones. So far, Human Headphones have had mixed reviews but there's no reason to think why this concept wouldn't get more popular as the technology evolves more.

In the next 10 years or so, we could also see earphones diminishing in size considerably to a point where they're almost invisible. Companies such as Third Skin are already working on nearly invisible wireless earbuds, but they're for boosting hearing. We could see the miniaturisation of earbuds in the commercial audio space in the future as well since there is a market of people that prefer the inconspicuous look. We also expect implantable headphones sometime in the future. One such concept is devised by designers James Auger and Jimmy Loizeau. The device they envision is designed to fit like a false tooth and can receive and transmit audio to and from the user using bone conduction. Currently, this is just a mind child of these two individuals but it wouldn't be far off to assume the proliferation of this kind of technology in 20 years, especially since bone conduction headphones already exist on the market. We could also see more modular headphones in the future like Apple's AirPods Max where the ear cups can be replaced if they were to get damaged. iFixit's teardown of the AirPods Max also showed that

the headband could be removed with just a SIM tool. Future headphones could come with even more modular or easily-swappable parts allowing for simpler repair jobs and relatively inexpensive replacement of modular/swappable parts that get damaged.

Augmented reality

According to a paper written by Leopold Bosankic, we will witness a decentralisation of the smartphone in the next decade. This means that the smartphone components will move from the smartphone to other gadgets, our environment, or even our bodies. As a result, we will rely on the smartphone's software more, but its hardware less. Augmented reality headsets are one such example where the screen moves from the smartphone to our heads. We are already seeing headphones with Wi-Fi built-in, so in a few years we may see more Wi-Fi enabled earbuds. Combine this with mesh Wi-Fi routers and you can possibly go longer distances away from your smartphone and still use its capabilities via the Wi-Fi enabled headset. These headphones and earbuds will allow us to stream music, fire up smart assistants, and much more, while on the go without your smartphone in close proximity.

Roy Choudhury, Professor of Electrical Engineering and Computer Science at the University of Illinois at Urbana-Champaign is on a mission with his team to transform today's earbuds into a whole new computing platform. He told Digital Trends, "Most of the wearable computing market has [so far] focused on devices that are worn on the lower part of the body, mostly in your pockets or on your wrists, maybe in your shoes. That means that you get to sense the lower part of the body, such as what you're doing with your hands, with your wrists, with your legs. But there's a lot of information that gets generated on the upper part of the body, mainly your head — such as listening, speaking, eating, facial emotions, potentially clues for medical-related information — that could be invaluable. The holy grail,